CONCEPTUAL RESEARCH NOTE | VERSION 0.1.6

Informational Cyclic Cosmology

A Conceptual Hypothesis of Relational Time, Algebra-Relative Entropy, and Aeonic Readability

Patrik William Pustejovsky

ORCID 0009-0008-1618-6619 | Concept DOI 10.5281/zenodo.21115416

Released 9 October 2026 | Open preprint

Status: speculative hypothesis, not a completed physical theory.

Revision note: version 0.1.6 adds a finite restricted-entropy stability bound and a bounded applicability review of prescribed observable-algebra transport. It also clarifies the limits of mass inaccessibility and cross-algebra gravitational entropy comparisons. Earlier selector and measurement results are unchanged. No physical algebra selector, cosmological entropy functional, or aeonic transition is derived.

ABSTRACT

This note explores a speculative cosmological interpretation in which an aeonic boundary changes the physically privileged local observable algebra of one globally conserved informational state, rather than restarting evolution in external time. The mechanism selecting that algebra is not provided. The proposal remains a conceptual research program without a distinct quantitative cosmological prediction; no new general theorem about factorization or observational entropy is claimed.

Finite studies delimit the proposal. A preliminary obstruction concerns a specific unrestricted Hamiltonian interaction-projection selector. Restricted Einstein-Cartan-inspired CAR examples do not evade the missing physical input: their three-cell robustness and four-cell EC-specificity tests fail. A distinct finite-time scrambling functional admits an internally certified entangled strict local minimum, but a product-state competitor has lower cost and better recovery in the stipulated operational task. This is not successful physical selection and does not contradict the interaction-cost obstruction.

A separate two-qubit illustration compares measurements on a fixed Bell state. The original reduced-state entropies are one and zero bits. A specified randomized Pauli protocol instead gives observational entropies of two and five-thirds bits, with an explicit white-noise and measurement-sharpness dependence. Its new measurement statistics are reproducible by local operations on both original qubits and classical communication, but not first-qubit-only access for arbitrary inputs at nonzero sharpness. Matching these statistics does not implement coherent control of the new subsystem. The comparison is not a temporal entropy decrease or a thermodynamic, gravitational, or cosmological reset.

At fixed finite factor dimension, a continuity bound controls restricted-state entropy changes under operator-norm algebra perturbations. Prescribed algebra transport supplies geometric comparison, not physical selection. Physical algebra selection, a justified cosmological entropy functional, relational transition dynamics, repeatability, and observational tests remain open. All mathematical and physical checks reported here are internal and LLM-assisted, not independent expert peer review.

Scope and Claimed Contribution

The mathematical ingredients used in this note are drawn from existing work on relational time, observable algebras, factorization-dependent entanglement, quantum mereology, observational entropy, and Einstein-Cartan fermion dynamics. The original finite-dimensional refactorization example is not presented as a new mathematical result. The selector-specific no-go argument added in version 0.1.2 and the restricted EC/CAR finite example and stress tests added in version 0.1.3 are presented for independent checking, not as established general theorems or as a derived cosmological mechanism. Version 0.1.5 adds a bounded finite-time counterexample and an operational elaboration of the original two-qubit illustration. These are different tasks with different inputs, not components already joined into a cosmological mechanism.

Version 0.1.6 adds finite stability and prescribed-transport comparisons using established tools; no novelty or independent expert certification is claimed.

The narrower claim explored here is that these ingredients may admit a distinct cosmological application. In this interpretation, successive aeons are not fundamentally successive global states. They are different locally readable descriptions associated with different physically privileged observable algebras and clock decompositions of one global state.

This is presently a research question, not an established mechanism. For the proposal to acquire physical content, it must explain what selects the local algebra, in what sense the selection constitutes an aeonic transition rather than a change of notation, and what observable consequence distinguishes it from existing cosmological models.

A non-exhaustive literature review found close prior art for the mathematical ingredients, cosmological observable-algebra embeddings, and relationally induced subsystem algebras. It did not identify the same aeonic interpretation, but that does not establish priority. The claimed contribution is therefore limited to the proposed synthesis and research question.

Results and Limits

Study	Result within its stated domain	What does not follow
Interaction-projection selector	Preliminary unrestricted strict-minimum obstruction	A no-go theorem for all selectors or ICC
Restricted EC/CAR examples	Finite competing minima; robustness and specificity failures	An EC-specific or scalable selector
Finite-time scrambling	Entangled strict local minimum, outperformed by a product competitor	Global preference or physical selection
Fixed measurement entropy	2 versus 5/3 bits under declared measurements	Thermodynamic or cosmological entropy reset
Measurement access	Exact local implementation with classical communication	Coherent subsystem control or a derived change of access
Finite algebra perturbations	Fixed-dimension restricted-entropy continuity bound	POVM or cosmological entropy stability without further assumptions
Prescribed algebra transport	Horizontal paths can realize a fixed-state entropy contrast	A selected path, physical transition, or thermodynamic arrow

The v0.1.5 reproducibility supplement separates analytic identities, exact-arithmetic certificates, numerical checks and assumed physical resources. None is presented as external peer review. The v0.1.6 supplement adds the bounded stability and transport analyses without reopening the closed selector searches.

1. Core Postulates

Foundational postulate: Primacy of the global informational structure

Fundamental reality is hypothesized to be a global informational structure. Time, space, particles, causality, and the thermodynamic arrow are treated as emergent features of how this structure is locally decomposed and read from within.

In the present ansatz, "informational structure" is not an additional substance or a numerical information measure. It is represented provisionally by a Hilbert space, a global observable algebra, a state on that algebra, and a Hamiltonian or constraint. The emergence claims are programmatic and are not derived in this note.

1. Global unitarity

Information is postulated never to be fundamentally destroyed. Apparent information loss is local, coarse-grained, or observer-dependent. In this version, black-hole evaporation must ultimately preserve information.

Because the proposal does not assume a fundamental external time, "global unitarity" does not mean evolution of the universe against an external clock. It means that the fundamental formulation contains no information-destroying non-unitary law and that any relational dynamics recovered within an aeon must be compatible with unitary quantum mechanics.

2. Entropy as loss of local readability

Within the hypothesis, growth of the relevant effective entropy does not mean destruction of the global state. It is interpreted as information becoming increasingly delocalized, correlated, or difficult to reconstruct relative to a specified local algebra and coarse-graining. This is not asserted as a definition of every thermodynamic or information-theoretic entropy.

3. Cyclicity as a property of reading, not temporal restart

The "end" and "beginning" of the universe are not necessarily two events separated by external time. They may be two candidate boundary descriptions of the same global informational structure. What appears as maximal disorder in one local decomposition may correspond to a differently embedded or refactored observable algebra for which a specified effective entropy is low.

4. Low effective entropy through changed distinguishability

The low entropy of a new cosmic phase is not produced by erasing information. It would require a differently embedded or refactored local observable algebra whose induced restricted state has low entropy under a specified entropy functional. Merely discarding observables or coarse-graining more strongly does not guarantee lower entropy and commonly increases observational entropy. This is

therefore not the claim that many microstates merely look identical under fixed macro-observables, which would normally correspond to high Boltzmann entropy. The required claim is narrower: the physically selected subsystem structure changes, and the same global state has high effective entropy in one embedded algebra but low effective entropy in the other. The selection and the entropy decrease are not derived here.

5. Cyclicity as a repeatability requirement

A single association between end-of-aeon and beginning-of-aeon descriptions is only a boundary proposal, not yet a cyclic cosmology. Calling the completed model cyclic requires an intrinsically selected, repeatable family of such associations together with conditional histories that run from a low-effective-entropy boundary toward a high-effective-entropy regime within each aeon. This does not require exact periodic recurrence of the same local state or algebra. Repeatability is a requirement of the hypothesis, not a result derived here. A fixed selection rule evaluated on identical global and relational input data would select the same physical algebra again; a sequence of distinct aeonic descriptions therefore requires intrinsically defined relational data that differ between the relevant conditional regimes.

2. Speculative Cosmological Extensions

The following conjectures are motivated by the informational framework but are not logical requirements of the aeonic boundary proposal. Each may be modified or falsified independently without by itself falsifying the central claim about local observable algebras.

2.1 Informational dark-sector conjecture

In this extension, the phenomena conventionally attributed to dark matter are conjectured not to arise from an additional particle species. They are instead the gravitationally accessible signature of correlations retained in the global informational state but inaccessible to the observable algebra of ordinary matter. Those correlations may encode structure associated with matter-like degrees of freedom in another aeonic decomposition while appearing in the current description only through gravity.

Because the central proposal has no fundamental external time, this should not yet be described as a signal propagating from a temporally earlier universe. It is instead a candidate residual pattern in the global state. Calling that pattern a wave would require a propagation equation, characteristic speed, and physical observables that have not been supplied.

At an effective level, one possible implementation would be

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{\text{visible}} + T_{\mu\nu}^{\text{info}}\right),$$

with total covariant conservation,

$$\nabla^{\mu} \left(T_{\mu\nu}^{\text{visible}} + T_{\mu\nu}^{\text{info}}\right) = 0.$$

Here $T_{\mu\nu}^{\text{info}}$ is only a placeholder. Calling it "information" does not explain why it gravitates. A completed extension must derive it from the global state and selected observable algebra, or replace it with a precise modification of the gravitational field equations. It must also reproduce the empirical successes normally attributed to cold dark matter.

2.2 Emergent rest-mass and scale conjecture

The algebraic boundary ansatz does not by itself require all massive particles to decay. The fate of rest mass becomes compulsory only if the aeonic identification also requires conformal scale invariance, as in conformal cyclic constructions. In that case, the extension must specify whether massive degrees of freedom disappear dynamically or become locally inaccessible under the selected new algebra while their information remains global. Local inaccessibility alone does not establish conformal scale invariance: mass scales may remain in the effective dynamics or gravitational backreaction. A conformal extension must show that those scales become irrelevant to the effective equations and observables used in the aeonic identification.

2.3 Dark energy as possible boundary data

Dark energy may drive extreme dilution and thereby contribute to the intrinsic boundary data λ used by the algebra selector. A dynamical dark-energy component might instead participate directly in selecting a new local description. Neither possibility is currently derived. If the cosmological constant remains positive and fixed, the extension must explain why an asymptotic de Sitter regime produces an aeonic algebra change rather than only continued expansion.

3. Algebraic Boundary Ansatz and Missing Selection Rule

For the present type-I ansatz, let $\mathcal{H}$ be a Hilbert space, $\mathcal{M} = \mathcal{B}(\mathcal{H})$ the global observable algebra, H a Hamiltonian or global constraint, and $\rho = |\Psi\rangle\langle\Psi|$ a pure stationary global state. In a Hamiltonian description stationarity means $[H, \rho] = 0$; in a constrained description it may instead take the form $H|\Psi\rangle = 0$. The expectation functional is

$$\omega(X) = \mathrm{Tr}(\rho X), \qquad X \in \mathcal{M}.$$

Purity and stationarity are assumptions of this version; neither follows from unitarity alone. The proposed aeonic identification is not represented by replacing one global state with another:

$$|\Psi_{\mathrm{old}}\rangle \to |\Psi_{\mathrm{new}}\rangle$$

At the schematic level, it instead associates two embedded local descriptions of the same state:

$$(\rho, \iota_{\mathrm{old}}(\mathcal{A}_{\mathrm{old}})) \;\; \mathcal{R} \;\; (\rho, \iota_{\mathrm{new}}(\mathcal{A}_{\mathrm{new}}))$$

Here:

```
rho = assumed pure, stationary global state
A_old = abstract local observable algebra in the old aeon
A_new = abstract local observable algebra in the new aeon
iota = embedding of an abstract local algebra into the global algebra
```

The proposal concerns the embedding and factorization of local observables inside the same global algebra, not a replacement of the underlying state. Two local algebras may be abstractly isomorphic while being embedded differently in the global algebra and therefore carrying different restricted states. The $\mathcal{B}(\mathcal{H})$ notation is schematic and type-I; it is not asserted to describe local algebras in continuum QFT.

At the present stage, let $\mathcal{R}$ be an unspecified relation contained in the product of the relevant embedding spaces. The symbol T_n labels one candidate pair in that relation:

$$\iota_n^{\infty} : \mathcal{A}_n^{\infty} \hookrightarrow \mathcal{M}, \qquad \iota_{n+1}^{0} : \mathcal{A}_{n+1}^{0} \hookrightarrow \mathcal{M},$$

$$T_n := \left(\iota_n^{\infty}, \iota_{n+1}^{0}\right) \in \mathcal{R}.$$

Here:

```
A_n^infty = candidate algebra assigned to the end-boundary regime of aeon n
A_{n+1}^0 = candidate algebra assigned to the start-boundary regime of aeon n+1
M = B(H) = schematic type-I global algebra
iota = embedding of a local algebra into the global algebra
R = proposed boundary relation, not yet a physical law
T_n = one candidate pair of related embeddings
```

The superscripts ∞ and 0 are mnemonic boundary labels in this version. They do not denote values of an external time parameter or limits already defined in a topology on states, algebras, or embeddings. If a completed model interprets either boundary as a genuine asymptotic limit, it must specify an intrinsic relational parameter, the relevant topology, and a convergence criterion. The same qualification applies to λ_n^{∞} and λ_{n+1}^{0} below.

This notation does not define T_n as a unitary operator on $\mathcal{H}$, a completely positive trace-preserving map on states, a homomorphism from the old algebra to the new one, or evolution in an external time parameter. It records the proposed aeonic association and nothing more. The index n labels the candidate aeonic descriptions; it is not an external time coordinate. A full theory must replace $\mathcal{R}$ with an intrinsic rule that selects the two embeddings and relates their operational meanings.

One candidate pair T_n describes only one proposed boundary. A cyclic model would require an intrinsically selected family

$$\{T_n\}_{n \in \mathcal{I}}, \qquad T_n = \left(\iota_n^{\infty}, \iota_{n+1}^{0}\right) \in \mathcal{R},$$

for an aeonic index set $\mathcal{I}$, plus compatible relational histories within the aeons. The index remains bookkeeping for relationally ordered descriptions, not a fundamental external time. Neither the existence nor the repeatability of this family is established here.

The global state is assumed to be the same on both sides:

$$\rho_{n+1} = \rho_n = \rho.$$

For any embedding $\iota : \mathcal{A} \hookrightarrow \mathcal{M}$, the induced state on the abstract local algebra is

$$\omega_{\iota}(a) := \omega(\iota(a)), \qquad a \in \mathcal{A}.$$

Because ρ is assumed pure, its global von Neumann entropy is

$$S_{\mathrm{vN}}(\rho) = 0.$$

The hypothesis requires, rather than derives, a target inequality of the form

$$S_{\mathrm{eff}}(\omega_{\iota_n^{\infty}}; \mathcal{A}_n^{\infty}) \gg S_{\mathrm{eff}}(\omega_{\iota_{n+1}^{0}}; \mathcal{A}_{n+1}^{0}).$$

Thus the beginning of a new aeon is not caused by information erasure. It is hypothesized to correspond to a change in which distinctions are locally readable. The equations above do not explain why that change occurs or establish the inequality.

Here S_{eff} should not be read as ordinary global von Neumann entropy, which is invariant under unitary evolution. It is a placeholder for an observational or subalgebra-relative entropy and must be specified together with the embedded algebra and coarse-graining. In a finite-dimensional bipartite matrix factor, the restricted state can be represented by a reduced density matrix. For a pure global state, its von Neumann entropy is the bipartite entanglement entropy S_{ent} used in Section 5; it is not automatically the cosmological S_{eff}. More general subalgebras, algebras with centers, and type-III algebras require different constructions. A specified quantum channel could model an operational coarse-graining only after its input algebra, output algebra, and physical meaning are fixed; an arbitrary channel is not a definition of local restriction. None of these entropy constructions defines the relation $\mathcal{R}$ or selects a candidate pair T_n. The current version does not choose a unique entropy construction for the full cosmological case.

Required form of a physical selection rule

Let $\mathfrak{E}_\lambda$ denote a physically admissible family of embeddings of candidate local algebras into $\mathcal{M}$, and let $\sim_{\mathrm{phys}}$ identify embeddings that represent the same operational physics. The label λ must be intrinsic: for example, relational clock data, asymptotic observables, or boundary conditions. It must not be an external time coordinate introduced only to make the aeons consecutive.

A completed model would need a selector of schematic form:

$$\big[\iota_\star(\lambda)\big] \in \operatorname*{arg\,min}_{[\iota]\in\mathfrak{E}_\lambda/\sim_{\mathrm{phys}}} \mathcal{F}([\iota];\rho,H,\lambda),$$

where H may be a Hamiltonian or global constraint and $\mathcal{F}$ is an operationally defined criterion that must be invariant under $\sim_{\mathrm{phys}}$. Existing quantum-mereology programs suggest criteria based on quasiclassicality, limited entanglement growth, pointer-state stability, locality, or minimal scrambling. The selector must not be defined merely by choosing whichever factorization makes ρ low entropy; that would build the desired conclusion into the premise.

For fixed ρ, H, and λ, a selector with a unique physical minimum would return the same equivalence class of embeddings. Distinct aeonic descriptions therefore require a family of intrinsic relational data, such as λ_n^∞ and λ_{n+1}^0, for which schematically

$$\big[\iota_n^\infty\big] = \big[\iota_\star(\lambda_n^\infty)\big], \qquad \big[\iota_{n+1}^0\big] = \big[\iota_\star(\lambda_{n+1}^0)\big].$$

These data must be derived from relational observables, records, conditional states, or boundary structure already encoded in the global theory. They cannot be supplied by the aeon index n, an external time variable, or the desired low-entropy conclusion. Because the clock decomposition is itself selected, a completed model may need a joint consistency condition for the embedding, clock, and relational boundary data rather than treating λ as an independent input. No such derivation or joint condition is provided here. If the selector instead has degenerate physical minima, the theory must derive a branch-selection or consistency rule; degeneracy alone does not generate an ordered aeonic sequence.

For physical content, the selector must identify a unique or physically equivalent class of embeddings, remain stable under small perturbations, and distinguish operationally meaningful observables from global unitary relabelings. The hypothesis does not yet supply such a functional or prove that its end-of-aeon and beginning-of-aeon solutions exist.

A tested selector and a preliminary entanglement no-go result

Version 0.1.2 investigates one deliberately simple Hamiltonian-only criterion. This is a diagnostic test of the selection problem, not an additional postulate. On a finite bipartite Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$, let a unitary U represent a candidate factorization and write

$$K_U = U^\dagger H U.$$

Let P_{loc} be the Hilbert-Schmidt orthogonal projection onto Kronecker-sum Hamiltonians $A \otimes I_B + I_A \otimes B$. Explicitly,

$$P_{\mathrm{loc}}(K) = \frac{1}{d_B}\mathrm{Tr}_B(K) \otimes I_B + I_A \otimes \frac{1}{d_A}\mathrm{Tr}_A(K) - \frac{\mathrm{Tr}\, K}{d_A d_B} I.$$

For a non-scalar H, define the normalized interaction cost

$$C_H([U]) = \frac{\|(I - P_{\mathrm{loc}})K_U\|_2^2}{\left\|H - \frac{\mathrm{Tr}\, H}{d_A d_B} I\right\|_2^2}.$$

This functional asks which candidate factorizations make the fixed Hamiltonian as close as possible to non-interacting. It does not include the target entropy and is therefore non-circular in that limited sense.

This is also the finite factor-algebra Gaussian minimal-scrambling criterion, up to normalization. Set $d = d_A d_B$ and $\mathcal{A}_U = U(\mathcal{B}(\mathcal{H}_A) \otimes I_B)U^\dagger$, with commutant $\mathcal{A}_U'$. In units $\hbar = 1$, equations (14) and (16) of Zanardi et al. (2024) give

$$\tau_s^{-2}(H, \mathcal{A}_U) = \frac{\|(I - P_{\mathcal{A}_U + \mathcal{A}_U'})H\|_2^2}{d},$$
$$C_H([U]) = \frac{d\,\tau_s^{-2}(H, \mathcal{A}_U)}{\|H - (\mathrm{Tr}\, H/d)I\|_2^2}.$$

where $P_{\mathcal{A}_U + \mathcal{A}_U'}$ projects onto the sum of the two operator subspaces, not their generated algebra. For fixed non-scalar H, the positive multiplier is independent of U: minimizing the squared Gaussian rate gives the same minima and local stability on the same candidate domain. This is direct prior art and operational motivation for the cost, not a different selector that evades the argument below. The bounded finite-time follow-up later in this section uses a different functional and a declared time-window protocol; its physical timing justification remains absent. The displayed Gaussian identification is not assumed for charge-sector algebras or continuum QFT.

The preliminary result is negative. Assume that ρ is a rank-one projector onto an isolated, nondegenerate eigenvalue of H, and quotient candidate factorizations by local basis changes, factor exchange where applicable, and global symmetries preserving both H and ρ. Then the current argument gives

> At every strict local minimum $[U]$ of C_H, the state $U^\dagger \rho U$ is a product state, and hence its bipartite entanglement entropy is zero.

The key steps are short. At a smooth critical point,

$$[K_U, P_{\mathrm{loc}}(K_U)] = 0.$$

The eigenvector of K_U associated with the selected nondegenerate eigenvalue is therefore also an eigenvector of the local operator $L = P_{\mathrm{loc}}(K_U) = A \otimes I + I \otimes B$. If its eigenvalue as an eigenvector of this Kronecker sum is nondegenerate, the eigenvector is a product vector. An entangled eigenvector consequently requires a degenerate eigenspace of L. Writing $K_U = L + J$, with $P_{\mathrm{loc}}(J) = 0$, and rotating by any unitary R inside that degenerate eigenspace gives

$$\left\|(I - P_{\mathrm{loc}})(R^\dagger K_U R)\right\|_2^2 = \|J\|_2^2 - \left\|P_{\mathrm{loc}}(R^\dagger J R)\right\|_2^2 \leq \|J\|_2^2.$$

To show that this is a genuine quotient direction, define the reduced-state purity

$$Q_\rho(U) = \mathrm{Tr}\left[\left(\mathrm{Tr}_B(U^\dagger \rho U)\right)^2\right].$$

This quantity is invariant under local basis changes, factor exchange for a pure state, and global symmetries preserving ρ. It therefore descends to the stated quotient. The degenerate eigenspace of L contains a product eigenvector $|\phi\rangle$. A unitary R_1 supported in that eigenspace can be chosen so that $R_1^\dagger |\psi\rangle = |\phi\rangle$, and it can be joined to the identity by an analytic path $R(t)$ inside the same block. Along this path the cost never exceeds its value at $t = 0$, while $Q_\rho(UR(t))$ is a nonconstant real-analytic function changing from a value below one to one. Consequently every neighborhood of $[U]$ contains a quotient-inequivalent point of no greater cost, which excludes a strict local minimum. This purity-based quotient argument is the step for which independent mathematical review is most important.

Exact two-qubit calculations, explicit degenerate and resonant examples, and numerical 2×3 searches support the argument. The tested degenerate examples display bipartite-entanglement contrast along flat families that fail strictness and robustness. This is not a general exclusion of strict minima for degenerate energies: state-changing rotations must also be shown inequivalent under the full quotient. Details and reproducible scripts are supplied in `feedback/general-no-go-analysis.md`, `feedback/two-qubit-preliminary-analysis.md`, and the `research` directory. The numerics are supporting checks, not a proof.

If independently confirmed, this result retires C_H as a solution to the specified finite-dimensional type-I entanglement-contrast problem under these assumptions. It does not determine whether algebras selected by this or another criterion could exhibit a contrast in a separately defined observational or subalgebra-relative S_{eff}. The Gaussian scrambling criterion inherits the same obstruction on this domain by the identity above. Genuinely different criteria, such as a specified finite-time scrambling or quasi-classicality functional, selectors acting on a family of observables, field-theoretic algebras, mixed or nonstationary states, and the central cosmological proposal require separate analysis.

Bounded finite-time follow-up

Version 0.1.5 tests a genuinely different finite-horizon functional. For a candidate embedding U, let $W_{U,t} = e^{-itU^\dagger H U}$, set $\hbar = 1$, and define the channel

$$\mathcal{E}_{U,t}(X) = \mathrm{Tr}_B\left[W_{U,t}(X \otimes I_B/d_B)W_{U,t}^\dagger\right].$$

Let $J_{U,t}$ be its trace-one Choi state, ordered as output tensor reference. The algebraic scrambling quantity and its finite-time average are

$$G_U(t) = 1 - \mathrm{Tr}(J_{U,t}^2), \qquad F_T([U]) = \frac{1}{T}\int_0^T G_U(t)\,dt.$$

This channel interpretation is established prior art (Styliaris, Anand & Zanardi), not a new entropy definition. Its operational use presupposes preparation of arbitrary A probes, a reset maximally mixed B, and specified timing. The Choi purity is not the reduced-state entropy of the selected stationary ground state. T is fixed independently of the candidate algebra and the desired entropy.

For $H = (XX + 2YY + 3ZZ)/\sqrt{14}$ and T = 8, the reference algebra has a quotient-strict local minimum with the nondegenerate singlet ground state $|\Psi^-\rangle$. Here PQ denotes $P \otimes Q$. Exact stationarity and rational enclosures of all six normal-slice Hessian eigenvalues give $\kappa_i > 7/100$ in the normalization of the technical derivation. This internally checked counterexample excludes a universal product-state strict-local-minimum claim for this finite-time functional. It is not a counterexample to the original cost-specific argument.

Crucially, the entangled local minimum is not a global minimizer. A product-eigenbasis embedding of the same H and selected state gives

$$F_8(\mathcal{A}_{\mathrm{Bell}}) \simeq 0.552031754, \qquad F_8(\mathcal{A}_{\mathrm{prod}}) \simeq 0.227611851.$$

The exact-arithmetic certificate bounds their difference strictly above 3/10. This particular branch therefore fails any near-global tolerance at most 0.3; neither a global optimizer nor a physically metastable algebra has been found. Its simple energies have repeated gaps. Finite-time local strictness does not contradict infinite-time global-minimizer results under nonresonance assumptions (Andreadakis, Dallas & Zanardi).

The operational comparison is also unfavorable to this Bell candidate. For the stated unital channel, put $P = \operatorname{Tr} J^2$ and let f_* denote optimal entanglement fidelity with an A-only CPTP decoder and maximally mixed input. The adjoint decoder and a spectral bound give $P \le f_* \le \sqrt{P}$, a specialization of transpose recovery (Barnum & Knill). For uniformly sampled, recorded time and a decoder allowed to depend on that time, the supplement certifies average optimal fidelities below 0.670 for the Bell candidate and above 0.772 for its product competitor. Unrecorded time defines a different channel and task; these bounds do not supply a cosmological clock or window.

This bounded follow-up is closed without promoting the branch to an ICC selector. Unitary transport generated by the same H stays in its centralizer orbit and thus in the same stipulated factorization class. A physical preference principle, timing prescription and transition between inequivalent classes remain additional inputs. The argument does not exclude other physically motivated selectors, and no such replacement is proposed here.

Restricted follow-up: an Einstein-Cartan-inspired CAR selector

Version 0.1.3 tests whether the negative result above still blocks the same cost after the admissible family is restricted by specified fermionic structure. Consider two spatial cells with four Dirac components per cell. At one-particle level require candidate transformations to preserve fermion number and normalize the internal matrix algebra

$$\mathfrak{D} = I_2 \otimes M_4(\mathbb{C}), \qquad V\mathfrak{D}V^\dagger = \mathfrak{D}.$$

Every such unitary has the form $V = W \otimes S$, where S is a common internal basis change that does not alter the cell split. The induced unordered cell decompositions form

$$U(2)/\left((U(1) \times U(1)) \rtimes S_2\right) \simeq \mathbb{RP}^2 \simeq S^2/\{\pm 1\}.$$

They are therefore represented by an unoriented Bloch axis $\mathbf{n}$. This restriction is a defining structural assumption of the finite model, not a consequence of the EC contact term; a completed theory would have to derive it independently.

For a massless one-link Dirac kinetic term and the normal-ordered axial-current contact channel induced when non-propagating Einstein-Cartan torsion is eliminated, the full-Fock interaction cost becomes

$$C(\mathbf{n}; r) = \frac{(1 - n_y^2) + 3r^2(1 - n_z^2)(3 + n_z^2)}{1 + 16r^2}, \qquad r = \frac{g}{t}.$$

For $1/\sqrt{12} < |r| < 1/\sqrt{6}$, it has two strict local endpoint minima, at the momentum split $\mathbf{n} = \pm\hat{\mathbf{y}}$ and site split $\mathbf{n} = \pm\hat{\mathbf{z}}$, which exchange global stability at $|r| = 1/3$. On the tested positive-coupling branch, $r = 1/3$, the Hamiltonian has an exactly certified isolated ground state and a numerically evaluated 1.80367-bit difference between its site- and momentum-factor entanglement entropies. The selector geometry depends only on r^2, but the stationary-state calculation is not asserted here for $r = -1/3$.

As a weighting gate, the calculation was repeated in the fixed four-particle sector rather than using the trace on the entire Fock space. In the full-Fock trace, sectors contribute through their sector traces and hence in proportion to their dimensions when expressed as normalized sector averages. The restricted cost is then

$$C_4(\mathbf{n}; r) = \frac{160(1 - n_y^2) + 24r^2(1 - n_z^2)(71 + 25n_z^2)}{160 + (15744/7)r^2}.$$

The two strict selector branches and finite barrier survive, with transition $|r_c| = \sqrt{20/213}$. On the tested positive branch $r = +\sqrt{20/213}$, the numerically isolated ground state remains in the four-particle sector and has a 2.00732-bit site-versus-momentum entanglement contrast. Appendix A states the construction, checks, and limitations, with reproducible code in the `research` directory.

This is not a counterexample to the unrestricted no-go argument: strictness is obtained only on a narrower candidate manifold from which the general descent directions have been removed. Its relevance therefore depends on whether the Dirac/CAR restriction follows from physical structure already present in a continuum theory. If it is imposed only to avoid the no-go directions, the escape is circular. The result establishes finite mathematical possibility under the stated restriction, not an aeonic transition, a cosmological entropy reset, or Einstein-Cartan cosmology.

A null control replacing the axial-current contact by a spin-independent onsite density contact reproduces the same two-minimum geometry. The finite transition should therefore be interpreted as generic competition between a hopping term that is local in momentum variables and an onsite interaction that is local in site variables, not as evidence for Einstein-Cartan dynamics. Conversely, the ground-state entropy contrast remains positive after imposing local-number or local-parity superselection, so it is not entirely a particle-number-fluctuation artifact. Appendix A gives the numerical controls and their limitations.

A first larger finite test replaces the two cells by three and admits the full restricted cell-mixing family $U(3)/(U(1)^3 \rtimes S_3)$. In the half-filled $N = 6$ sector, the open-chain kinetic endpoint has a negative Hessian direction and an explicit lower mixed factorization. The periodic ring retains two numerically strict endpoints, and a strengthened multi-start search found no lower candidate, but this is not a proof of global minimality. At three cells, changing the boundary also changes two bonds into three and changes the discrete-derivative spectrum at order one. The combined result therefore fails a boundary-

independent robustness gate. The periodic branch is recorded as a finite numerical observation, not as support for an EC-derived selector. Appendix A and the three-cell technical report give the construction, exact rank certificate, controls, and limitations.

A four-cell periodic ring then tests the full $U(4)/(U(1)^4 \rtimes S_4)$ cell-mixing family in the half-filled $N = 8$ sector. Both periodic endpoints are numerically strict. An explicit gauge-transported branch stays stationary and strict at the prescribed small twists $\pi/16$ and $\pi/8$, but loses positive curvature at $\pi/2$. This removes the narrow objection that the three-cell periodic observation was purely an odd-size effect, while still failing full periodic-to-antiperiodic persistence.

More decisively, a prospectively fixed spinor-blind density contact $Q_{\mathrm{dens}} = \sum_x N_x(N_x - 1)$ passes the same periodic and small-twist gates. The two contacts are not equivalent after fixed-sector centering, yet both obey the same transported residual law and retuned coexistence relation $g_*(\phi) = g_0 \cos(\phi/4)$. The finite behavior is therefore not an EC-specific signature. Appendix A, the matched-control report, and its separate mathematical and physics reviews state the result and limits.

In compact form:

```
same global state rho
+ old embedded algebra -> high specified effective entropy
+ new embedded algebra -> low specified effective entropy
```

4. Relational Clock Time, the Thermodynamic Arrow, and Clock Ambiguity

The Page-Wootters mechanism addresses how conditional dynamics may be encoded in a globally stationary state. It does not by itself explain why one temporal orientation is thermodynamically distinguished. Relational clock time and the thermodynamic arrow are therefore treated as separate problems in this note.

Idealized conditional dynamics

In the idealized construction, the accessible Hilbert space and observable algebra admit compatible clock/rest decompositions:

$$\mathcal{H} \simeq \mathcal{H}_C \otimes \mathcal{H}_R, \qquad \mathcal{A}_n \simeq \mathcal{A}_n^C \otimes \mathcal{A}_n^R.$$

The stationary global state must also satisfy an appropriate constraint. In the ideal non-interacting form,

$$H_{\mathrm{tot}}|\Psi\rangle = 0, \qquad H_{\mathrm{tot}} = H_C \otimes I_R + I_C \otimes H_R.$$

The zero eigenvalue is conventional; a stationary energy eigenvalue can be shifted to zero. A nonzero constrained state also requires compatible clock and rest energy support, so that contributing energy eigenvalues can sum to the chosen total eigenvalue. Interactions, imperfect clocks, and more general constraints require extensions of this ideal form.

Let the clock states be generated by H_C,

$$|t\rangle_C = e^{-iH_C t}|0\rangle_C, \qquad E_t = |t\rangle_C\langle t|.$$

For $\rho = |\Psi\rangle\langle\Psi|$, the state of the remaining degrees of freedom relative to the clock reading t is

$$\rho_R(t) = \frac{\mathrm{Tr}_C[(E_t \otimes I_R)\rho]}{p(t)}, \qquad p(t) = \mathrm{Tr}[(E_t \otimes I_R)\rho].$$

The formula applies where $p(t) > 0$. Each E_t is treated as a rank-one positive effect and, for normalized $|t\rangle$, an individual rank-one projection. A continuous family of such effects in a finite-dimensional clock is generally non-orthogonal and is not a projection-valued time observable; a normalized clock POVM requires an appropriate measure and normalization. Actual measurement updates additionally require a specified instrument. The conditional state above is instead a relative-state assignment and does not require an actual measurement of the clock. Under the ideal constraint and spectral-compatibility assumptions above, it satisfies, in units with $\hbar = 1$,

$$\frac{\partial \rho_R(t)}{\partial t} = -i[H_R, \rho_R(t)].$$

Thus t is an internal label for correlations between a selected clock and the rest, not a fundamental external parameter. The construction recovers reversible conditional dynamics; it does not yet select a preferred clock or a direction of time.

The separate arrow-of-time requirement

The ideal conditional evolution above is unitary. If $\rho_R(t)$ is pure, its von Neumann entropy remains zero. A thermodynamic arrow therefore requires additional structure: a physically selected coarse-graining or accessible subalgebra, a low-observational-entropy boundary condition, and conditional dynamics that produce records and typical entropy growth away from that boundary.

The Page-Wootters construction does not by itself identify either orientation of the clock label as the thermodynamic future. A thermodynamic arrow would orient the clock readings in the direction of typical entropy increase away from the selected low-entropy boundary.

Schematically, for a selected observational algebra $\mathcal{A}_R^{\mathrm{obs}}$, an arrow would require behavior such as

$$S_{\mathrm{obs}}(\rho_R(t_2) \mid \mathcal{A}_R^{\mathrm{obs}}) \gtrsim S_{\mathrm{obs}}(\rho_R(t_1) \mid \mathcal{A}_R^{\mathrm{obs}}), \qquad t_2 > t_1,$$

for typical conditional histories over an appropriate regime. This is not asserted as exact monotonicity: finite systems can fluctuate or recur. Here S_{obs} is a placeholder for a specified observational or coarse-grained entropy, not the global von Neumann entropy. The proposed low-effective-entropy restricted state relative to the new aeon's algebra could serve as a boundary condition only if that algebra is selected non-circularly and its conditional dynamics actually generate the required asymmetry. Refactorization alone does not do so.

Clock ambiguity

A stationary global state generally admits many candidate clock/rest decompositions. Different choices can encode different conditional histories and even different effective dynamical laws. This is the clock ambiguity. It is especially important here because the proposal already allows the physically privileged observable algebra to differ between aeons.

The status of the ambiguity is disputed. Marletto and Vedral argue that fixing the global Hamiltonian and requiring an ideal non-interacting clock can select a unique suitable decomposition. Recent work by Stoica argues that non-interaction alone is insufficient and that the physical meaning of the observables must also be supplied. This note does not assume that the dispute is settled.

The selector introduced in Section 3 must therefore select not only an embedded local algebra, but also a compatible clock subalgebra and rest algebra, and it must relate them to the fixed global Hamiltonian or constraint and to a physical record structure. It must not choose a clock merely because that clock reproduces the desired history. Across the aeonic boundary relation, a different selected clock may define a different relational time coordinate, but it does not by itself define a new thermodynamic arrow.

5. Finite-Dimensional Illustration

This section first illustrates an established kinematic fact: the entanglement and reduced-state entropy assigned to a pure state depend on the chosen subsystem factorization or observable algebra. Sections 5.2 and 5.3 then specify an observational-entropy comparison and the measurements that realize it. None models cosmology, derives an aeonic transition, or claims a new general theorem.

The operational role of observable algebras in selecting quantum subsystems was developed by Zanardi. Explicit constructions show that observables can induce a tensor-product structure in which an arbitrary pure state has a chosen Schmidt decomposition (Harshman & Ranade), and that a state may appear entangled or separable under different factorizations of its observable algebra (Thirring et al.). The calculation below is a two-qubit specialization of this known factorization dependence, included to make the proposed cosmological interpretation concrete.

5.1 Factorization-relative entanglement

For this illustration only, define the bipartite entanglement entropy S_{ent} as the von Neumann entropy of the reduced state obtained from the assumed pure global state. This quantity depends on the chosen tensor-product factorization. It is not identified here with the cosmological S_{eff} of Section 3 or the observational entropy used for the thermodynamic arrow in Section 4.

Let the global Hilbert space be finite-dimensional:

$$\mathcal{H} \simeq \mathbb{C}^4.$$

In the old aeon, choose a two-qubit factorization:

$$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B, \qquad \dim \mathcal{H}_A = \dim \mathcal{H}_B = 2.$$

Let the globally conserved state be a Bell state with respect to this factorization:

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|0_A 0_B\rangle + |1_A 1_B\rangle).$$

The old local observable algebra is:

$$\mathcal{A}_n^{\infty} = \mathcal{B}(\mathcal{H}_A) \otimes I_B.$$

An observer restricted to this algebra traces over the inaccessible subsystem B:

$$\rho_A = \mathrm{Tr}_B(|\Psi\rangle\langle\Psi|) = \frac{I_A}{2}.$$

The entanglement entropy relative to the old factorization is:

$$S_{\mathrm{ent}}(\rho \mid \mathcal{A}_n^{\infty}) = S(\rho_A) = -\mathrm{Tr}(\rho_A \log_2 \rho_A) = 1.$$

Thus, relative to the old factorization, the locally accessible state is maximally mixed.

Now choose a second factorization of the same global Hilbert space, intended only to represent the candidate local description assigned to a new aeon:

$$\mathcal{H} = \mathcal{H}_{A'} \otimes \mathcal{H}_{B'}.$$

Because the tensor-product structure is not fixed by the abstract four-dimensional Hilbert space alone, one may choose a new factorization in which the same vector $|\Psi\rangle$ is separable:

$$|\Psi\rangle = |0_{A'}\rangle \otimes |0_{B'}\rangle.$$

Explicitly, one may define a new product basis by setting:

$$|0_{A'}0_{B'}\rangle = \frac{1}{\sqrt{2}}(|0_A0_B\rangle + |1_A1_B\rangle),$$

$$|0_{A'}1_{B'}\rangle = \frac{1}{\sqrt{2}}(|0_A0_B\rangle - |1_A1_B\rangle),$$

$$|1_{A'}0_{B'}\rangle = |0_A1_B\rangle, \qquad |1_{A'}1_{B'}\rangle = |1_A0_B\rangle.$$

This defines a valid tensor-product structure $\mathcal{H}_{A'} \otimes \mathcal{H}_{B'}$ on the same abstract four-dimensional Hilbert space, with the original Bell state now appearing as a product state in the primed factorization.

Equivalently, let U be the global unitary that maps the original computational product basis, in order, to the four primed basis vectors above. Then

$$|\Psi\rangle = U|0_A0_B\rangle, \qquad \mathcal{A}^0_{n+1} = U\mathcal{A}^\infty_n U^\dagger.$$

This makes explicit that the two finite-dimensional factorizations are related by a global unitary. The illustration supplies no physical reason to privilege one embedding over the other.

The new local observable algebra is:

$$\mathcal{A}^0_{n+1} = \mathcal{B}(\mathcal{H}_{A'}) \otimes I_{B'}.$$

The reduced state seen by the new local observer is:

$$\rho_{A'} = \mathrm{Tr}_{B'}(|\Psi\rangle\langle\Psi|) = |0_{A'}\rangle\langle 0_{A'}|.$$

Therefore:

$$S_{\mathrm{ent}}(\rho \mid \mathcal{A}^0_{n+1}) = S(\rho_{A'}) = 0.$$

The assumed global state has not changed:

$$\rho_{n+1} = \rho_n = \rho.$$

The global von Neumann entropy remains zero throughout:

$$S_{\mathrm{vN}}(\rho) = 0.$$

The two descriptions differ only in their chosen local factorization and accessible observable algebra.

The values above are entanglement entropies of reduced density matrices. They are not calculations of Boltzmann entropy, observational entropy, horizon or gravitational entropy, or the entropy of a cosmological heat-death state. The illustration therefore does not establish the target inequality for S_{eff} in Section 3; it only demonstrates that one factorization-relative entropy can differ for the same pure vector.

The illustration reproduces the following standard factorization dependence:

```
same global pure state
+ old factorization
-> locally maximally mixed state

same global pure state
+ new factorization
-> locally pure state
```

The calculation does not show that a refactorization occurs, nor that either factorization is physically privileged. In finite-dimensional Hilbert space, an arbitrary refactorization can make entanglement entropy appear or disappear by construction. A physical theory must therefore provide a selection principle for the factorization and separately connect the selected algebra to a thermodynamically relevant entropy. One possible route is quantum mereology: the preferred factorization is not arbitrary, but selected by a Hamiltonian, locality, robustness of pointer states, or quasi-classical dynamics. The interaction-projection cost tested in Section 3 is one simple Hamiltonian-locality proxy, and its preliminary no-go result shows that this particular route does not supply the bipartite-entanglement contrast used in the finite-dimensional illustration. It does not settle the paper's distinct, still-unspecified cosmological S_{eff}. More sophisticated selectors still need independent, non-circular motivation. In the cosmological hypothesis, the missing task is to explain why extreme dilution, dark-energy dominance, or a boundary condition at an aeonic transition would naturally select the new algebra $\mathcal{A}^0_{n+1}$.

There is also a field-theoretic limitation. The illustration uses a finite-dimensional type-I Hilbert-space factorization, where partial traces and reduced density matrices are well-defined. Local algebras in non-gravitational relativistic quantum field theory are typically type-III von Neumann algebras, where a spatial tensor-product factorization and ordinary reduced density matrices are not available in the same way (Witten). Gravitationally dressed observable algebras need not have that type: a de Sitter static-patch construction yields a type-II_1 algebra with a corresponding entropy (Chandrasekaran et al.), while clock-defined algebras in semiclassical cosmological spacetimes can be type-II_∞ (Chen & Penington). A full extension must therefore identify the physically relevant algebraic class and define locality, entropy, and restriction without relying on naive partial tracing.

Comparing entropies of different gravitational algebras also requires a common, physically justified reference and compatible normalization and renormalization conventions. A type-II_1 factor has a canonical normalized trace, whereas a type-II_∞ factor's semifinite trace is defined only up to a positive rescaling, which shifts its state entropy by an additive constant (Chen & Penington). Where algebra entropy represents generalized entropy, the comparison must consistently include the horizon contribution and matter entropy. In the cited de Sitter static-patch construction, empty de Sitter space is the maximum-entropy state, not a low-entropy initial state (Chandrasekaran et al.). Thus dilution alone does not establish the proposed entropy contrast, and independently chosen entropy offsets cannot establish it either. This additional continuum requirement does not change the declared conventions or results of the finite examples.

5.2 A fixed measurement-relative entropy

Keep the state $\rho = |\Phi^+\rangle\langle\Phi^+|$ and the original U above, whose columns are $(\Phi^+, \Phi^-, 01, 10)$. This is a separate example from Section 3's singlet/Hamiltonian calculation. Neither the state nor its two embeddings is selected by that finite-time criterion.

For a specified POVM $\mathcal{C} = \{E_i\}$, use the established observational entropy, in bits,

$$S_{\mathcal{C}}(\rho) = -\sum_i p_i \log_2(p_i/V_i), \qquad p_i = \mathrm{Tr}(\rho E_i), \quad V_i = \mathrm{Tr}\, E_i.$$

All traces defining volume use the same global four-dimensional space; zero-probability terms vanish. This applies the definition of Buscemi, Schindler & Safranek. The protocol chooses one of X,Y,Z uniformly, measures its binary outcome $s = \pm 1$, and retains both axis and sign. It is not a serial measurement of three incompatible observables. With $W = I_4$ or U for the two embeddings,

$$E^{W,\eta}_{j,s} = \frac{1}{6} W[(I_2 + s\eta\sigma_j) \otimes I_2]W^\dagger, \qquad 0 \le \eta \le 1.$$

Each effect has volume 2/3. The axis weights, frame, retained records and volume convention are fixed before the comparison. For the stipulated white-noise family $\rho_p = (1-p)\rho + pI_4/4$, $0 \le p \le 1$, put $x = \eta(1-p)$ and let h_2 be the binary Shannon entropy. Direct evaluation gives

$$S_{\mathcal{C}_I}(\rho_p) = 2, \qquad S_{\mathcal{C}_U}(\rho_p) = \frac{5}{3} + \frac{1}{3} h_2\left(\frac{1+x}{2}\right).$$

At ideal purity and sharpness the contrast is 1/3 bit. It stays positive for $p < 1$ and $\eta > 0$ in this family, tending to zero at complete mixing or zero sharpness. This is not a general apparatus-noise or finite-sample robustness claim. X or Y alone yields two bits in both descriptions; Z alone yields two versus one at the ideal point. The definition is therefore explicitly measurement-relative, not fixed by the algebra alone.

This state-specific contrast does not establish a measurement ordering valid for every input, or superior performance on every information-processing task. For general POVMs, classical-postprocessing, measured-relative-entropy and observational-entropy orderings are inequivalent: Teixido-Bonfill, Schindler & Safranek (2025). Accordingly, "readability" here refers only to the declared state family and measurement protocol, not universal information access or a physical selection principle.

Quantity at the ideal point	Old description	New description
Global von Neumann entropy	0 bits	0 bits
Reduced-state entropy, equal to entanglement entropy for this pure input	1 bit	0 bits
Fixed XYZ observational entropy	2 bits	5/3 bits
Cosmological S_{eff}	Not supplied	Not supplied

The inaccessible factor contributes one bit through global trace volumes; subtracting it would change the defined quantity. The six effects span the embedded qubit algebra for $\eta > 0$, but tomography of its marginal is not the same entropy calculation. For mixed inputs reduced-state entropy is not an entanglement measure. A generic POVM's observational entropy is also not the von Neumann entropy of its Petz reconstruction or postmeasurement state.

Simultaneous conjugation of state and effects preserves all probabilities and volumes. Here the effects change relative to a fixed state. Consequently the lower value is a comparison of measurement descriptions, not a trajectory of decreasing entropy. The detailed formulas, controls and fixed-grid checks are in the operational supplement.

5.3 Physical access and closure

The new observables have the explicit original-factor form

$$X' = \frac{X \otimes I + Z \otimes X}{\sqrt{2}}, \quad Y' = \frac{I \otimes Y + Y \otimes Z}{\sqrt{2}}, \quad Z' = Z \otimes Z.$$

Their outcome statistics can be implemented by local operations and classical communication (LOCC), assuming both original qubits are addressable:

Setting	Local implementation
X'	Measure X on B, send result b to A, then measure $(X + bZ)/\sqrt{2}$ on A.
Y'	Measure Y on A, send result a to B, then measure $(Y + aZ)/\sqrt{2}$ on B.
Z'	Measure Z on both and report the product of signs.

Retain only the chosen axis and final sign. The branch effects in the supplement sum to the target POVM effects, including the axis probability 1/3. This operator identity holds for every input, not only the Bell state. An independent sign flip of probability $(1 - \eta)/2$ realizes the same unsharp effects. No inter-qubit gate, quantum communication between the original subsystems or additional shared entangled ancilla is needed for these statistics. This is a sufficient protocol, not an optimum in communication, time, energy or apparatus cost.

A-only access is insufficient for $\eta > 0$: $|00\rangle$ and $|01\rangle$ have the same A marginal but different parity statistics. At $\eta = 0$ the outputs are state-independent. The construction thus exposes an added access assumption; ICC has not derived why both-factor access becomes available. Equal outcome counts and trace volumes do not establish equal physical resources.

These measurements do not implement arbitrary coherent operations on the new subsystem. On $|{+}{+}\rangle$, ideal coherence-preserving parity measurement has a Bell state as its normalized even-parity output, whereas separate local Z readouts give $(|00\rangle\langle 00| + |11\rangle\langle 11|)/2$. The outcome probabilities agree, but the instruments differ. Unassisted LOCC cannot realize the ideal instrument with its quantum outputs on the original A and B, because that would create entanglement from a product input; see Chitambar et al. for the operational classes.

Intermediate local results also exist even when omitted from the final record. Retaining them defines a refined POVM, whose entropy need not equal the six-outcome value. Omitting a record is not a modeled thermodynamic erasure or restoration of coherence. No physical entropy reset, selected algebra, field-theoretic extension or cosmological transition follows. The finite-example sequence is closed at this operational boundary.

5.4 Stability under finite algebra perturbations

A separate robustness question concerns the restricted von Neumann entropy used in Section 5.1, not the POVM entropy in Section 5.2. Let $\mathcal{H} = \mathbb{C}^d \otimes \mathbb{C}^m$, $d \geq 2$, and let $A, B \subset \mathcal{B}(\mathcal{H})$ be unital factors of the same type M_d and multiplicity m. For any fixed density matrix ρ, choose embeddings $\iota_A(a) = W_A(a \otimes I_m)W_A^\dagger$ and define

$$\sigma_A = \mathrm{Tr}_m(W_A^\dagger \rho W_A), \qquad S_\rho(A) = -\mathrm{Tr}_d(\sigma_A \log_2 \sigma_A),$$

with the analogous definition for B. This uses ordinary local matrix trace and bits and is independent of local basis. For mixed global states it is not an entanglement measure.

Let $\delta = d_{\mathrm{KK}}(A, B)$ be the two-sided operator-norm Hausdorff distance between their entire closed unit balls, following Kadison and Kastler (1972). Define

$$\omega_d(x) = h_2(t) + t \log_2(d-1), \qquad t = \min\{x, 1 - 1/d\},$$

where h_2 is binary entropy. Then the direct finite-factor bound is

$$|S_\rho(A) - S_\rho(B)| \le \omega_d(\delta).$$

To see this, the trace-preserving conditional expectation E_B obeys $\|a - E_B(a)\| \le 2\delta$ on A's unit ball. The pulled-back map $\Phi = \iota_B^{-1} E_B \iota_A$ is completely positive, unital and trace-preserving because the multiplicities agree. Its trace adjoint has the same properties. For $\tau_A = \Phi^*(\sigma_B)$, trace-norm duality gives $\frac{1}{2}\|\sigma_A - \tau_A\|_1 \le \delta$, while bistochasticity gives $S(\tau_A) \ge S(\sigma_B)$. Applying Audenaert's continuity inequality and then interchanging A and B proves the bound. The cap in ω_d is essential when substituting an upper bound on trace distance.

For example, $d = 2$ and $\delta \le 0.01$ limit the change to approximately 0.080793 bits. A two-endpoint comparison has the sum of the two corresponding error bounds. These are conditional guarantees: a finite sample of observables does not certify the required unit-ball distance. The proof does not select an algebra, require a pure or stationary state, or construct a physical operation changing access.

The full derivation and checks also give an explicit growing-dimension sequence whose algebra distances tend to zero while its entropy gaps tend to one bit. Thus this estimate cannot become dimension-independent without additional restrictions. It is not a type-III, gravitational, or continuum entropy theorem, and it does not control the entropy of an unspecified measurement protocol. No new universal Kadison-Kastler stability theorem is assumed or certified here.

6. Relation to Existing Frameworks

LambdaCDM

The standard model remains the observational baseline. Planck 2018 found strong consistency with six-parameter flat LambdaCDM and no compelling evidence for extensions: Planck 2018. This hypothesis differs by treating the cosmic endpoint as informationally continuous with an effective new beginning.

DESI and dark energy

DESI DR2 BAO measurements are well described by flat LambdaCDM on their own. When combined with CMB and supernova data, however, they favor a time-varying dark-energy equation of state at a significance that depends on the supernova sample, with unknown systematics still a possible qualification: DESI DR2. This matters because the fate of expansion determines whether dilution, boundary formation, or another transition mechanism is possible.

Penrose CCC

Penrose's conformal cyclic cosmology joins the future conformal infinity of one aeon to the conformally stretched Big Bang of the next across a crossover surface: Meissner & Penrose. Its defining transition is conformal-geometric; it does not identify successive aeons as different locally readable observable alge-

bras of one fixed global quantum state. Proposed CMB Hawking-point evidence remains contested. An et al. claim a detection, while Jow and Scott and Bodnia et al. report no statistically significant evidence: An et al., Jow & Scott, Bodnia et al..

Exactly periodic quantum cyclic cosmology

Carroll, Diachenko, and Dulani propose an exactly periodic cosmology in a finite-dimensional Hilbert space and discuss quantum mereology as a still preliminary route to a semiclassical interpretation. Their mechanism relies on unitary state evolution, commensurable energy gaps, and a distinguished low-entropy excursion associated with a Crunch/Bounce/Big-Bang phase: Carroll, Diachenko & Dulani.

The present hypothesis shares the use of finite-dimensional quantum mereology as a possible route to emergent subsystem structure, but proposes a different mechanism. It does not identify the cycle primarily with periodic evolution of the global state. Instead, it asks whether an effectively new aeon can arise through a change in the locally accessible observable algebra or factorization of a globally conserved state. Whether that distinction can be made physically meaningful, rather than merely redescribing the same dynamics, is an open question.

Black-hole information

Recent Page-curve and island results support the possibility that black-hole evaporation can preserve information in quantum gravity models: Almheiri et al., Raju review.

Emergent spacetime and holography

AdS/CFT supplies a concrete gauge/gravity duality, the Ryu-Takayanagi proposal relates boundary entanglement entropy to bulk minimal-surface area, and Van Raamsdonk argues that bulk connectivity is tied to entanglement: Maldacena, Ryu-Takayanagi, Van Raamsdonk. These results motivate information-theoretic approaches to geometry in specific holographic settings; they do not establish the present cosmological proposal.

Relational time and clock ambiguity

The hypothesis is related to Page-Wootters-style relational time, where a stationary constrained state can encode conditional dynamics relative to an internal clock: Page & Wootters, Moreva et al., Marletto & Vedral. The ideal construction requires a compatible clock/rest decomposition and constraint; interactions or mixed global states can modify the resulting conditional dynamics: Rijavec.

The non-uniqueness of the clock choice can underdetermine the effective history and laws: Albrecht & Iglesias. Whether ideal non-interaction removes this ambiguity is contested by a recent analysis that instead requires physical meaning to be assigned to the observables: Stoica. None of these constructions derives a thermodynamic arrow from relational time alone. An extended finite-dimensional Page-Wootters model obtains an informational arrow only after adding inaccessible auxiliary degrees of freedom and effective noisy dynamics: Jesni Shamsul Shaari. This provides an example of the additional coarse-graining needed, not a derivation of a macroscopic cosmological arrow from the clock mechanism itself.

Relational and cosmological observable algebras

Quantum-reference-frame work demonstrates that different relational perspectives can induce different subsystem observable algebras and therefore different notions of subsystem locality and entanglement (Ahmad et al.). This is close kinematic prior art for the present proposal, but it does not identify those perspectives with successive aeons or supply an entropy-reset mechanism.

Observable-algebra embeddings have also been used to extract a cosmological sector from full loop quantum gravity (Koslowski). More recent constructions use physical clocks or observer dressing to define type-II gravitational observable algebras in de Sitter and other cosmological settings (Chandrasekaran et al., Chen & Penington). These results are relevant to the missing definitions of physical accessibility and effective entropy, but none provides the aeonic boundary relation or repeatable selector proposed here.

Factorization-relative entanglement

The finite-dimensional illustration uses the established fact that subsystem structure, entanglement, and reduced-state entropy depend on the observable algebra or tensor-product structure. This operational and algebraic relativity is developed in Zanardi, Harshman & Ranade, and Thirring et al.. The present note does not extend those mathematical results; it asks whether they admit a physically selected aeonic interpretation.

Transport of observable algebras

Zanardi (2026) develops a canonical connection and holonomic transport on finite observable-algebra orbits of fixed block structure. This supplies transport along a specified path, not the path itself. The bounded applicability review exhibits a horizontal two-qubit path with a one-to-zero-bit restricted-entropy contrast for a fixed Bell state. Co-transporting the state and algebra instead preserves that entropy. The path and stopping point are prescribed, not selected by independent physical data. Geometric reparameterization invariance does not supply a relational clock or remove the control and gap requirements of an adiabatic implementation. This is conditional geometric prior art, not an ICC selector, cosmological transition, or extension to continuum algebras.

Quantum mereology and observational entropy

Quantum mereology studies how preferred subsystem decompositions might be selected rather than assumed in advance. Carroll and Singh propose selecting quasi-classical factorizations by limiting entanglement growth and internal spreading: Carroll & Singh. Operational quantum mereology instead represents generalized subsystem structures by an observable algebra and its commutant, and uses minimal algebraic scrambling as a dynamical criterion: Zanardi et al.. Its Gaussian criterion is equivalent to the interaction-projection cost on the finite bipartite factor-algebra domain, as shown above; it is not an independent replacement for that tested selector. A related toy-universe model jointly selects Hamiltonian structure, subsystem decomposition, and a low-entropy state through a variational principle: Shokrian Zini, Brown & Freedman.

None of these results supplies the aeonic boundary relation proposed here, but they constrain what a non-arbitrary selector $\mathcal{F}$ could mean. The proposed cosmological entropy remains unspecified. Section 5.2 uses an explicit POVM-relative instance of established observational entropy, without identifying it with that cosmological quantity or with global von Neumann entropy: Safranek, Deutsch & Aguirre, Buscemi, Schindler & Safranek.

For specified accessible algebras, the recent preprint by Rignon-Bret & Elouard (2026) develops entropy accounting based on maximum-entropy completions and specified microscopic dynamics. This algebra-dependent construction is not automatically the general-POVM entropy used in Section 5.2. It provides conditional thermodynamic context, not a derivation of ICC's proposed boundary relation. Changing an algebraic description alone does not establish a physical transition, thermodynamic entropy reduction, or cosmological cycle; the selection mechanism, operational dynamics and cosmological entropy remain unresolved here.

In quantum field theory, entanglement is more naturally treated as a property of local observable algebras rather than only as a property of states on tensor-product factors: Witten.

Constraints on the informational dark-sector extension

The extension's strongest empirical challenge is dark matter phenomenology. Observations such as the Bullet Cluster show gravitational lensing mass separated from the dominant baryonic X-ray gas, strongly favoring some unseen gravitating component: Clowe, Randall & Markevitch. Any informational-gravitational replacement must reproduce galaxy rotation curves, lensing, CMB acoustic peaks, BAO, cluster collisions, and structure growth at least as well as cold dark matter. The CMB spectra provide an independent precision constraint: a replacement must reproduce the successful base-LambdaCDM fit to the acoustic structure, not only late-time galaxy dynamics (Planck 2018).

7. Failure Criteria and Missing Physical Content

The central proposal's first vulnerability is the algebra-selection rule. In this version, $\mathcal{R}$ is only an unspecified boundary relation and T_n is one candidate pair of embeddings in that relation, not a quantum channel or time-evolution map. The proposal must define the admissible family $\mathfrak{C}_\lambda$, the intrinsic data λ, and a selector $\mathcal{F}$ that chooses a stable physical equivalence class without assuming the desired entropy decrease. It must also explain how relationally derived data such as λ_n^∞ and λ_{n+1}^0 can differ while the global state and constraint remain fixed; otherwise repeated evaluation selects the same algebra and no aeonic change occurs.

Over the unrestricted finite bipartite factorization space, the simple interaction-projection selector tested in Section 3 appears to fail the entanglement-contrast test: under the stated assumptions, its strict local minima cannot support an entangled isolated stationary eigenstate. This result should be independently checked, especially its quotient-space step based on reduced-state purity. The restricted EC/CAR calculation does not refute it; it removes the relevant descent directions by limiting the admissible manifold. That finite construction passes the full-Fock and fixed-$N = 4$ tests, but its physical relevance now depends on deriving the restriction independently in a continuum theory. Neither result addresses a separately specified observational or subalgebra-relative S_{eff}.

The separate finite-time example shows why the cost-specific obstruction cannot simply be generalized to every finite-time strict local minimum. It does not supply physical selection: its entangled branch is outperformed by the stated product competitor. Likewise, Section 5 supplies a concrete measurement-relative entropy and an all-input measurement implementation, not an independently motivated selection or change of access. These tasks remain separate; their partial results do not jointly constitute a transition law.

The boundary labels are also not yet mathematical limits. Any asymptotic interpretation of the superscripts ∞ and 0 requires a relational parameter, topology, and convergence law; otherwise they remain labels for the two sides of the proposed association.

The second vulnerability is physical equivalence. If the old and new embeddings differ only by a global unitary relabeling, with no intrinsic operational criterion distinguishing them, the proposal has no physical content beyond a change of description.

The third vulnerability is temporal orientation. Page-Wootters conditioning can recover relational dynamics, but the proposal must separately derive a low-entropy boundary, typical observational-entropy growth, and stable records. It must also show that the algebra selector chooses a physically meaningful clock/rest decomposition rather than one engineered to encode the desired history.

The fourth vulnerability is the field-theoretic and gravitational extension. The finite-dimensional illustration uses type-I tensor-product factorization and partial traces. A realistic version must identify the appropriate local or gravitational observable algebras, which may involve type-III algebras in local QFT or type-II algebras after gravitational dressing, and must define effective entropy without relying on the naive finite-dimensional reduced density matrix.

The fifth vulnerability is observational uniqueness. The hypothesis must predict something different from LambdaCDM, Penrose CCC, exactly periodic finite-dimensional quantum cyclic cosmology, and existing emergent-gravity models.

The sixth vulnerability is cyclicity itself. One admissible boundary pair does not establish a cycle. The proposal must show that its intrinsic selection rule can generate a repeatable family of compatible aeonic boundaries and within-aeon relational histories without inserting each new low-entropy description by hand.

Extension-specific failure conditions

The informational dark-sector extension fails in its present form if it cannot derive a covariantly consistent gravitational source and reproduce CMB acoustic peaks, BAO, lensing, cluster collisions, halo phenomenology, and structure growth. A direct particle detection that accounts for the dark-matter phenomenology without a residual informational component would falsify the extension's stated non-particle interpretation, not automatically the central aeonic proposal.

Stable massive particles are fatal only to versions that require a conformal loss of scale. They do not by themselves falsify the more general algebraic boundary relation.

The dark-energy extension remains unsupported until a fixed cosmological constant or a dynamical dark-energy field is shown to provide intrinsic boundary data or a selection mechanism for the new observable algebra.

8. Working Summary

This is not yet a complete physical theory. It is a structured hypothesis:

> The universe may be one globally conserved informational structure. Time, entropy, matter, and cosmic history are local readings of that structure. A cosmic cycle would not be a restart in external time, but a change in local readability: the same global information that appears maximally disordered at the end of one description may induce a low-effective-entropy state on a differently embedded or refactored local observable algebra at the beginning of another.

The proposed contribution is this aeonic interpretation of a change in the physically privileged local observable algebra. The note does not claim a new general factorization theorem, and it does not yet derive an aeonic transition from a Hamiltonian, boundary condition, or other dynamical principle. One simple Hamiltonian-only selector fails the stated unrestricted finite-dimensional entanglement test, while an Einstein-Cartan-inspired restriction of its admissible CAR family yields a bounded finite example with strict competing minima and stationary entanglement contrast at two cells. Its first three-cell extension fails boundary-independent robustness. A four-cell periodic extension has a locally robust small-twist branch, but a matched non-axial density contact reproduces the primary gate and the same branch-cost law. The final bounded classification shows that every nonzero contact in the declared translation-invariant onsite quartic class shares that law. The construction is therefore retained only as a bounded example of generic kinetic-versus-onsite competition rather than a scalable or EC-specific selector candidate. The restriction is not yet derived from continuum physics, and none of these results decides whether a physically defined cosmological S_{eff} could exhibit the proposed contrast.

The finite-time follow-up admits an entangled strict local minimum but does not make it globally preferred, operationally superior in the specified recovery task, or physically selected. Independently, the two-qubit measurement protocol gives a fixed-state observational-entropy contrast of 1/3 bit and an explicit local implementation with classical communication. Neither comparison is a physical entropy-reducing process. The finite studies clarify operational meaning and limitations; they do not derive algebra selection, relational transition dynamics, or cosmological cycling. This version closes these finite examples rather than extending their scope.

To become a physical theory, the hypothesis needs:

1. a rigorous definition of the admissible observable algebras and their embeddings, including a relational definition of the boundary regimes and a topology if genuine limits are claimed,
2. a precise relation $\mathcal{R}$ and a criterion selecting the candidate pair T_n independently of the desired entropy conclusion,
3. an intrinsic, non-circular rule that jointly determines the physically privileged algebra and the relational data λ_n that can distinguish successive conditional regimes,
4. a demonstration that the selected construction is repeatable across aeonic boundaries rather than specifying only one pair,
5. a physically justified cosmological definition of effective entropy `S_eff` and its target contrast, rather than an inference from either toy entanglement entropy or the stipulated finite measurement protocol,
6. a field-theoretic version that can handle local operator algebras beyond the finite-dimensional illustration,
7. a compatible relational-clock construction and a separate derivation of the thermodynamic arrow,
8. at least one observational prediction that differs from LambdaCDM, Penrose CCC, exactly periodic quantum cyclic cosmology, and emergent-gravity models.

The optional cosmological extensions additionally require a derivation of $T^{\text{info}}_{\mu\nu}$ or another precise gravitational modification, a clear statement of whether conformal scale loss is required, and a mechanism by which dark energy supplies boundary data rather than merely cosmic dilution.

Appendix A. Restricted Einstein-Cartan-Inspired CAR Selector

This appendix summarizes a finite follow-up to the selector problem in Section 3. It is supplementary to the central ICC hypothesis. The complete derivation, additional perturbation checks, and executable calculations are supplied in the companion derivation, the full-Fock calculation, the exact spectral certificate, the fixed-sector gate, the adversarial controls, and the superselection controls. The first larger finite stress test is supplied in the three-cell report and its executable calculation. The ranked internal critique is the v0.1.3 adversarial review. The proof-level follow-up is the v0.1.3 mathematical audit. The three-cell increment received separate physics and mathematics reviews before integration. The four-cell extension is supplied in the finite gate, boundary-phase report, and matched-control report, with separate mathematical and physics reviews. The full onsite class is treated in the analytic closure, its reproducibility script, and the separate mathematical review.

A.1 Finite fermionic model

Take two spatial cells and four Dirac components per cell:

$$\mathfrak{h} = \mathbb{C}^2_{\text{cell}} \otimes \mathbb{C}^4_{\text{Dirac}}, \qquad \mathcal{H} = \mathcal{F}(\mathfrak{h}), \qquad \dim \mathcal{H} = 2^8 = 256.$$

With the link spacing absorbed into the coefficients, define

$$H(t, g) = tK + gQ,$$

$$K = -i\left(\psi_1^\dagger \alpha^1 \psi_2 - \psi_2^\dagger \alpha^1 \psi_1\right),$$

and

$$Q = \sum_{x=1}^{2}\left[:\left(\psi_x^\dagger \gamma^5 \psi_x\right)^2: - \sum_{i=1}^{3} :\left(\psi_x^\dagger \alpha^i \gamma^5 \psi_x\right)^2:\right].$$

The contact term is the finite-mode axial-axial channel corresponding to $J_5^\mu J_\mu^5$. Eliminating algebraic, non-propagating torsion in Einstein-Cartan fermion theory produces a local four-fermion interaction that includes this channel. Its sign and coefficient depend on conventions and on possible non-minimal couplings (Diakonov, Tumanov & Vladimirov). Only g^2 enters the selector branch diagram. For comparison, Khanapurkar et al. derive a non-relativistic limit directly from the Einstein-Cartan-Dirac field equations. The present finite-mode truncation is neither a discretized solution of those equations nor such a controlled limit.

A microscopic Einstein-Cartan cosmology with torsion-induced effective four-fermion interactions can produce a bounce (Lucat & Prokopec), but that construction does not derive the algebra selector used here.

A.2 Restricted candidate factorizations

Arbitrary factorizations of the 256-dimensional Fock space belong to the unrestricted setting of the no-go argument. The number-preserving Fock-space lifts $\Gamma(V)$ of one-particle $U(8)$ transformations form a proper subgroup of $U(256)$, not the unrestricted domain. We restrict to these one-particle transformations and further require normalization of the internal Dirac matrix algebra

$$\mathfrak{D} = I_2 \otimes M_4(\mathbb{C}), \qquad V\mathfrak{D}V^\dagger = \mathfrak{D}.$$

Because every automorphism of $M_4(\mathbb{C})$ is inner, every such unitary has the form

$$V = W \otimes S, \qquad W \in U(2), \quad S \in U(4).$$

The common S acts only as an internal basis change inside each candidate cell, so it does not alter the factorization. The induced unordered cell decompositions form

$$U(2)/\left((U(1)\times U(1)) \rtimes S_2\right) \simeq \mathbb{RP}^2.$$

Equivalently, the candidate split is represented by

$$\mathbf{n} = (n_x, n_y, n_z) \in S^2, \qquad \mathbf{n} \sim -\mathbf{n}.$$

Fermionic mode factors use a graded tensor product. The finite implementation fixes a Jordan-Wigner ordering and works with parity-even Hamiltonians and number-preserving candidate unitaries. The normalizer condition is the defining restriction of this proof of concept. Its physical derivation is not supplied.

A.3 Full-Fock selector and branch diagram

Lift W to Fock space and project the transformed Hamiltonian onto one-factor Kronecker-sum Hamiltonians using the Hilbert-Schmidt projection from Section 3. The denominator in Section 3 removes the scalar component of the Hamiltonian. Here $\operatorname{Tr} K = 0$ and $\operatorname{Tr} Q/256 = 4$. Define $\widetilde{Q} = Q - 4I$. Direct CAR traces give

$$\|K\|_2^2 = 512, \qquad \|\widetilde{Q}\|_2^2 = 8192, \qquad \langle K, \widetilde{Q}\rangle_{\mathrm{HS}} = 0,$$

$$\|K_{\mathrm{int}}(\mathbf{n})\|_2^2 = 512(1-n_y^2),$$

$$\|Q_{\mathrm{int}}(\mathbf{n})\|_2^2 = 1536(1-n_z^2)(3+n_z^2),$$

and a vanishing residual cross term. Therefore, with $r = g/t$,

$$C(\mathbf{n}; r) = \frac{(1-n_y^2) + 3r^2(1-n_z^2)(3+n_z^2)}{1+16r^2}.$$

For fixed n_z, minimization sets $n_x = 0$. Writing $u = n_z^2$, the numerator is

$$N(u; r) = u + 3r^2(1-u)(3+u),$$

a concave quadratic for $r \neq 0$. Hence every global minimum is an endpoint class. The momentum factorization $\mathbf{n} = \pm\hat{\mathbf{y}}$ is global for $|r| < 1/3$, the site factorization $\mathbf{n} = \pm\hat{\mathbf{z}}$ is global for $|r| > 1/3$, and they are co-global at $|r| = 1/3$. Both remain locally stable in the interval

$$\frac{1}{\sqrt{12}} < |r| < \frac{1}{\sqrt{6}}.$$

At the transition,

$$C_{\min} = \frac{9}{25}, \qquad C_{\text{barrier}} = \frac{39}{100}, \qquad \Delta C = \frac{3}{100}.$$

Along the shortest yz path $\mathbf{n}(\alpha) = (0, \sin 2\alpha, \cos 2\alpha)$, four equal fermionic beam splitters give the finite Grassmannian line element

$$ds^2_{\text{CAR}} = 4\, d\alpha^2, \qquad d_{\text{CAR}}(\text{site}, \text{momentum}) = \frac{\pi}{2}.$$

This metric and the finite barrier permit a circuit-penalized update model, but they do not derive a unique circuit action, transition rate, or relational clock law. The geometric language follows the general circuit-geometry program (Nielsen) and its fermionic developments (Hackl & Myers); no complexity functional from those works is claimed to select this cosmological model.

A.4 Fixed-particle-number gate

The full-Fock Hilbert-Schmidt inner product uses the trace on the entire Fock space:

$$\mathrm{Tr}_{\mathcal{H}}(X) = \sum_{N=0}^{8} \mathrm{Tr}_{\mathcal{H}_N}(P_N X P_N).$$

It therefore weights orthonormal Fock-space basis states equally, not particle-number sectors equally. In a normalized-trace decomposition, sector N carries weight $\binom{8}{N}/256$. The primary weighting gate restricts instead to

$$\mathcal{H}_{N=4} = \bigoplus_{k=0}^{4} \left(\wedge^k \mathfrak{h}_+\right) \otimes \left(\wedge^{4-k} \mathfrak{h}_-\right), \qquad \dim \mathcal{H}_{N=4} = 70.$$

Let P_4 be the total-number projector and let $N_\pm$ be the two local number operators. Fix the local Hamiltonian subspace before selecting H or $\mathbf{n}$:

$$\mathfrak{L}_4 = \left\{ P_4 (A_+ \widehat{\otimes} I + I \widehat{\otimes} A_-) P_4 : [A_+, N_+] = [A_-, N_-] = 0 \right\}.$$

It has dimension 135. In the charge block with dimensions $d_k = \binom{4}{k}$ and d_{4-k}, the orthogonal projection is

$$\Pi_k(X) = \frac{\mathrm{Tr}_- X}{d_{4-k}} \otimes I + I \otimes \frac{\mathrm{Tr}_+ X}{d_k} - \frac{\mathrm{Tr}\, X}{d_k d_{4-k}} I,$$

and off-diagonal charge blocks project to zero. This handles the common particle-number center without counting its parametrization kernel as an extra local direction.

In this sector $\mathrm{Tr}\, K = 0$ and $\mathrm{Tr}\, Q/70 = 24/7$. With $\widetilde{Q}_4 = Q - (24/7) I_{70}$, the centered sector traces give

$$\|K\|^2_{2,4} = 160, \qquad \|\widetilde{Q}_4\|^2_{2,4} = \frac{15744}{7}, \qquad \langle K, \widetilde{Q}_4 \rangle_{2,4} = 0,$$

$$\|K_{\text{int}}(\mathbf{n})\|^2_{2,4} = 160(1 - n_y^2),$$

$$\|Q_{\text{int}}(\mathbf{n})\|^2_{2,4} = 24(1 - n_z^2)(71 + 25 n_z^2),$$

again with zero residual cross term. The normalized fixed-sector selector is

$$C_4(\mathbf{n}; r) = \frac{160(1 - n_y^2) + 24 r^2 (1 - n_z^2)(71 + 25 n_z^2)}{160 + (15744/7) r^2}.$$

It retains the same two strict branches. Their exchange and coexistence window are

$$|r_c| = \sqrt{\frac{20}{213}} \simeq 0.3064257065,$$

$$\sqrt{\frac{5}{72}} < |r| < \sqrt{\frac{10}{69}}.$$

At the transition,

$$C_{4,\min} = \frac{497}{1153}, \qquad C_{4,\text{barrier}} = \frac{2163}{4612}, \qquad \Delta C_4 = \frac{175}{4612}.$$

Thus the two-branch result is not confined to the unrestricted full-Fock weighting. Gibbs and other state-weighted norms have not been tested.

A.5 Stationarity and certificate boundaries

At the positive-coupling full-Fock co-global point $H_c = K + Q/3$, numerical diagonalization gives

$$E_0 \simeq -4.11560796875, \qquad E_1 - E_0 \simeq 0.620514177337.$$

The supplied exact certificate constructs $3H_c$ directly from Gaussian-integer CAR data. Conserved particle-number, chirality, and link-spin labels split it into 65 blocks of maximum dimension 18. Exact rational LDL^{T} inertia and the spectral factor

$$p(\lambda) = \lambda^4 - 20\lambda^3 - 144\lambda^2 + 3008\lambda - 1792$$

prove that the ground state is simple and

$$E_1 - E_0 > \frac{923}{1500}.$$

Its mode-entanglement entropies are evaluated numerically:

$$S_{\text{site}} \simeq 3.01251822001 \text{ bits}, \qquad S_{\text{mom}} \simeq 1.20884898928 \text{ bits},$$

so $\Delta S_{\text{mode}} \simeq 1.80366923073$ bits. At the fixed-sector transition $r = \sqrt{20/213}$, the numerically isolated ground state remains in $N = 4$ and has

$$E_1 - E_0 \simeq 0.678109892996, \qquad \Delta S_{\text{mode}} \simeq 2.00731779928 \text{ bits}.$$

Because these ground states have fixed total particle number, their reduced states are block diagonal in local particle number. Resolving the reduced state into sectors allowed by local fermionic superselection reduces but does not remove this contrast. The reported quantity is the probability-weighted within-sector entropy; it excludes the Shannon entropy of the sector label. At the positive-coupling full-Fock exchange point, the site-minus-momentum contrast is numerically 0.698734 bits under local-number superselection and 0.995665 bits under local-parity superselection. At the positive fixed-$N = 4$ exchange point the corresponding values are 0.800293 and 1.162493 bits. These are operationally stricter fermionic pure-state diagnostics, not a general mixed-state entanglement measure or thermodynamic entropy. The local-number construction follows the standard particle-entanglement decomposition (Wiseman and Vaccaro); fermionic subsystems and parity superselection are reviewed by Szalay et al..

A 121-point scan across the fixed-sector bistability interval found a minimum sampled gap of 0.5359112 and a minimum sampled entropy contrast of 1.4856425 bits. The fixed-sector spectral and entropy claims are numerical, not interval-certified.

A.6 Three-cell finite-size stress test

The first size extension uses three cells, four Dirac modes per cell, and the half-filled sector

$$\mathcal{H}_{3,6} = \bigwedge^{6} \mathbb{C}^{12}, \qquad \dim \mathcal{H}_{3,6} = \binom{12}{6} = 924.$$

The admissible cell factorizations form $U(3)/(U(1)^3 \rtimes S_3)$ and have six continuous tangent directions. The fixed-sector one-cell Hamiltonian space has 210 matrix-unit generators. Two relations identify the three copies of the identity and a third is $N_1 + N_2 + N_3 = 6I$. Row reduction of the integer Gram matrix modulo the prime 1,000,003 gives rank 207; together with those three exact relations, this certifies the local-subspace dimension over $\mathbb{Q}$.

At each boundary condition the positive coupling is tuned to make the site and kinetic-eigenmode endpoint costs equal. The primary numerical result is:

Quantity	Open chain	Periodic ring
g_c	0.3501691221	0.4129483210
Normalized endpoint cost	0.4529794648	0.4717832957
Minimum site Hessian eigenvalue	0.1222247	0.4866812
Minimum kinetic Hessian eigenvalue	**-0.0465277**	0.2433404
Lowest optimized cost found	**0.4413803**	0.4717832957
Ground-space multiplicity	**3**	1
Gap above the ground space	0.5688714	0.2033920
Mode-entropy contrast	not evidential	1.7508546 bits
Local-number-SSR contrast	not evidential	0.6380137 bits
Local-parity-SSR contrast	not evidential	0.9787055 bits

The open kinetic endpoint is not a local minimum. Direct motion along its negative Hessian eigenvector lowers the cost, independently of optimizer convergence. The periodic endpoints are numerically strict; 32 Haar samples and optimization from eight random starts found no lower point, but do not prove global minimality on the flag manifold.

The normalized endpoint-Hessian pattern is the same at the tested fillings $N = 4, 5, 6, 7, 8$: the open kinetic endpoint fails and both periodic endpoints are locally strict. Periodic even fillings $N = 4, 6, 8$ have isolated positive-contrast ground states; the odd fillings have fourfold numerical ground-space degeneracy and their arbitrary eigenvector entropies are not used. A non-axial onsite density contact also produces strict competing endpoints, so the result is not specific evidence for torsion dynamics.

The boundary comparison is not a bulk-limit test. At $L = 3$, the open chain has two links while the periodic ring has three, and the periodic central-difference kinetic spectrum is $\{-\sqrt{3}, 0, +\sqrt{3}\}$. Both the graph and the naive-derivative spectrum therefore change at order one. The combined result is classified as a boundary-independent robustness failure, with the periodic branch retained only as a restricted finite numerical observation. No conclusion about larger size follows from the three-cell comparison alone.

A.7 Four-cell phase and matched-control gate

The even-size follow-up uses

$$\mathcal{H}_{4,8} = \bigwedge^{8} \mathbb{C}^{16}, \qquad \dim \mathcal{H}_{4,8} = \binom{16}{8} = 12{,}870,$$

and the 12-dimensional quotient $U(4)/(U(1)^4 \rtimes S_4)$. Exact combinatorial partial traces evaluate the centered Hilbert-Schmidt selector without constructing a dense $12{,}870 \times 12{,}870$ Hamiltonian. The implementation is checked against dense low-filling matrices, sparse half-filled occupation-mask actions, projection overlaps, traces, norms, quotient invariance, and finite-difference step convergence.

At periodic boundary conditions, the kinetic operator has a two-dimensional zero eigenspace. A fixed grid-plus-refinement rule resolves its basis freedom using the interaction selector before the endpoint coupling is set. At the equal-endpoint axial coupling $g_0 = 0.4298279139$, the site and resolved periodic representatives have minimum quotient-Hessian eigenvalues 0.177740 and 0.122137. A prescribed gauge transport

$$U_\phi = G(\phi) U_0, \qquad G_{xx}(\phi) = e^{ix\phi/4},$$

is stationary across the 17-point grid. It remains strict at $\pi/16$ and $\pi/8$, then first fails the grid at $\pi/2$ with minimum Hessian -0.010441. Thus the small-twist gate passes while the complete periodic-to-antiperiodic gate fails.

The prospectively selected comparison replaces the axial contact by the spinor-blind onsite density interaction $Q_{\mathrm{dens}} = \sum_x N_x(N_x - 1)$ and changes nothing else in the gate. The control is reconstructed exactly on every $N = 8$ occupation mask. After fixed-sector centering, its cosine with the axial contact is 0.2119995760 and it retains relative residual 0.9772697579 after the best axial rescaling, so it is not an affine copy under the selector.

At its equal-endpoint coupling $g_0 = 1.2984286349$, the density control passes the same periodic, $\pi/16$, and $\pi/8$ tests. It remains strict at $\pi/2$ and first loses positive curvature at $9\pi/16$. For both interactions the audited residuals satisfy

$$R_{\mathrm{site}} = (A, 0, 0), \qquad R_{\mathrm{branch}} = \left(A \sin^2 \frac{\phi}{4}, B, 0\right), \qquad g_0^2 B = A,$$

to errors below 7.0×10^{-10}. The corresponding retuned equal-cost root is $g_*(\phi) = g_0 \cos(\phi/4)$ for both contacts. Finite seeded searches at the first two nonzero twists find no lower candidate but do not prove global minimality.

The remaining onsite question admits a bounded analytic closure. Write the most general translation-invariant, number-preserving onsite quartic contact as $Q_q = \sum_x Q_x(q)$, with $q = q^\dagger \in \mathrm{Herm}(\Lambda^2 \mathbb{C}^4)$. This is a 36-real-dimensional space. If (β, χ) parameterizes the periodic kinetic zero-mode basis and $x = \sin^2\beta \cos^2\chi$, exact fixed-sector contractions give

$$B_q(\beta, \chi) = B_q(0,0) - p(q)x - r(q)x^2.$$

Under $\mathrm{Herm}(\Lambda^2 \mathbb{C}^4) = \mathbf{1} \oplus \mathbf{15} \oplus \mathbf{20}$, the spectra of p are $231/4$, $231/2$, and 231, while those of r are $693/4$, $539/4$, and $231/2$ on the respective summands. Both forms are positive definite. Every nonzero contact therefore makes the declared periodic resolver choose $x = 1$, including $\beta = \pi/2$. At that balanced representative, the symmetric-hopping residual has squared Hilbert-Schmidt norm A, and its residual cross contraction with every Q_q vanishes. Hence

$$R_{\mathrm{branch}}(\phi) = \left(A \sin^2 \frac{\phi}{4}, B(q), 0\right)$$

for the complete nonzero interaction class just defined. This is a theorem only inside the fixed four-cell model and resolver; it does not cover disordered or inter-cell contacts, other fillings or norms, or a continuum limit.

The prospective EC-specificity gate therefore fails. Here "non-axial" means only that the control is not the axial-current contraction used for the minimal EC-inspired channel; it does not assert that no generalized torsion effective theory can contain density-like interactions. The result tests one finite selector and does not simulate EC field dynamics.

The complete artifacts are the boundary-phase report, matched-control report, reproducibility script, and the separate mathematical and physics reviews. The full onsite classification is supplied in the closure report, closure script, and its separate mathematical review.

A.8 Interpretation and stop condition

The two-cell calculation demonstrates only the following finite statement: on a specified Dirac-algebra-normalizing, number-preserving CAR manifold, an Einstein-Cartan-inspired Hamiltonian supplies a non-entropic interaction cost with two strict minima, a finite barrier, and an isolated stationary state with factorization-dependent mode entanglement. The structure survives one physically motivated charge-sector weighting. The three-cell result shows that this statement is not boundary-independent at the first change in lattice size. The four-cell result supplies a locally robust periodic branch but shows that its mechanism is reproduced by an independent non-axial onsite control. The final bounded classification strengthens this from a two-contact observation to every nonzero interaction in the declared translation-invariant onsite quartic class.

Two adversarial controls narrow that statement. Direct traces in all fixed number sectors retain the two-minimum geometry for $2 \leq N \leq 6$, but the sector ground state at the equal-endpoint coupling is numerically degenerate for $N = 2, 3, 5, 6$; only $N = 4$ passes the isolated-state gate. More importantly, the non-axial contact $Q_{\mathrm{dens}} = \sum_x N_x(N_x - 1)$ produces the same qualitative selector geometry and a positive stationary contrast. Hence the finite geometry is generic kinetic-versus-onsite locality competition. The axial-current channel supplies one realization but no EC-specific explanatory evidence. The two-cell controls are reproduced by the adversarial-control calculation and the superselection calculation. The four-cell matched comparison strengthens this stop condition by applying the same branch and phase protocol to both interactions. The onsite closure shows that testing additional contacts from that same finite class cannot restore specificity.

It does not derive the admissible manifold from continuum field equations, establish validity at the Planckian coupling where the contact term competes with the kinetic term, extend the construction to type-III local QFT algebras, define cosmological observational entropy, produce a Page-Wootters boundary transition, demonstrate repeatable aeons, or predict a cosmological observable. It also does not show persistence at arbitrary cell count or that Einstein-Cartan dynamics selects this factorization family. These are required before the finite result can support the central ICC proposal.

References

- Ahmad, S. A., Galley, T. D., Hoehn, P. A., Lock, M. P. E., and Smith, A. R. H. (2022). Quantum Relativity of Subsystems. *Physical Review Letters* 128, 170401.
- Albrecht, A., and Iglesias, A. (2008). The clock ambiguity and the emergence of physical laws. *Physical Review D* 77, 063506.
- Almheiri, A., Mahajan, R., Maldacena, J., and Zhao, Y. (2020). The Page curve of Hawking radiation from semiclassical geometry. *Journal of High Energy Physics* 2020, 149.
- An, D., Meissner, K. A., Nurowski, P., and Penrose, R. (2018). Apparent evidence for Hawking points in the CMB Sky. arXiv:1808.01740.
- Andreadakis, F., Dallas, E., and Zanardi, P. (2024). Long-time Quantum Scrambling and Generalized Tensor Product Structures. *Physical Review A* 109, 052424.
- Audenaert, K. M. R. (2007). A sharp continuity estimate for the von Neumann entropy. *Journal of Physics A: Mathematical and Theoretical* 40, 8127-8136.
- Barnum, H., and Knill, E. (2002). Reversing quantum dynamics with near-optimal quantum and classical fidelity. *Journal of Mathematical Physics* 43, 2097-2106. Revised arXiv version, 2026.
- Bodnia, E., Isenbaev, V., Colburn, K., Swearngin, J., and Bouwmeester, D. (2022). The quest for CMB signatures of Conformal Cyclic Cosmology. arXiv:2208.06021.
- Buscemi, F., Schindler, J., and Safranek, D. (2023). Observational entropy, coarse quantum states, and Petz recovery: information-theoretic properties and bounds. *New Journal of Physics* 25, 053002.
- Carroll, S. M., Diachenko, N., and Dulani, S. (2026). Toward a Phenomenologically Acceptable Quantum Cyclic Universe. PhilSci-Archive preprint.
- Carroll, S. M., and Singh, A. (2021). Quantum Mereology: Factorizing Hilbert Space into Subsystems with Quasi-Classical Dynamics. *Physical Review A* 103, 022213.
- Chandrasekaran, V., Longo, R., Penington, G., and Witten, E. (2023). An Algebra of Observables for de Sitter Space. *Journal of High Energy Physics* 2023(02), 082.
- Chen, C.-H., and Penington, G. (2024). A clock is just a way to tell the time: gravitational algebras in cosmological spacetimes. arXiv:2406.02116.
- Chitambar, E., Leung, D., Mancinska, L., Ozols, M., and Winter, A. (2014). Everything You Always Wanted to Know About LOCC (But Were Afraid to Ask). *Communications in Mathematical Physics* 328, 303-326.
- Clowe, D., Randall, S. W., and Markevitch, M. (2007). Catching a bullet: direct evidence for the existence of dark matter. *Nuclear Physics B - Proceedings Supplements* 173, 28-31.
- DESI Collaboration (2025). DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints. *Physical Review D* 112, 083515.
- Diakonov, D., Tumanov, A. G., and Vladimirov, A. A. (2011). Low-energy general relativity with torsion: a systematic derivative expansion. *Physical Review D* 84, 124042.
- Hackl, L., and Myers, R. C. (2018). Circuit complexity for free fermions. *Journal of High Energy Physics* 2018(07), 139.
- Harshman, N. L., and Ranade, K. S. (2011). Observables can be tailored to change the entanglement of any pure state. *Physical Review A* 84, 012303.

- Jow, D. L., and Scott, D. (2020). Re-evaluating evidence for Hawking points in the CMB. *Journal of Cosmology and Astroparticle Physics* 2020(03), 021.
- Kadison, R. V., and Kastler, D. (1972). Perturbations of von Neumann Algebras. I. Stability of Type. *American Journal of Mathematics* 94, 38-54.
- Khanapurkar, S., Pradhan, A., Dhruv, A., and Singh, T. P. (2018). Non-relativistic limit of Einstein-Cartan-Dirac equations. *Physical Review D* 98, 104027.
- Koslowski, T. A. (2007). A Cosmological Sector in Loop Quantum Gravity. arXiv:0711.1098.
- Lucat, S., and Prokopec, T. (2017). Cosmological singularities and bounce in Cartan-Einstein theory. *Journal of Cosmology and Astroparticle Physics* 2017(10), 047.
- Maldacena, J. M. (1998). The Large N Limit of Superconformal Field Theories and Supergravity. *Advances in Theoretical and Mathematical Physics* 2, 231-252.
- Marletto, C., and Vedral, V. (2017). Evolution without evolution, and without ambiguities. *Physical Review D* 95, 043510.
- Meissner, K. A., and Penrose, R. (2025). The Physics of Conformal Cyclic Cosmology. arXiv:2503.24263.
- Moreva, E., et al. (2014). Time from quantum entanglement: an experimental illustration. *Physical Review A* 89, 052122.
- Nielsen, M. A. (2006). A geometric approach to quantum circuit lower bounds. *Quantum Information & Computation* 6, 213-262.
- Page, D. N., and Wootters, W. K. (1983). Evolution without evolution: Dynamics described by stationary observables. *Physical Review D* 27, 2885-2892.
- Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics* 641, A6.
- Raju, S. (2020). Lessons from the Information Paradox. arXiv:2012.05770.
- Rijavec, S. (2023). Robustness of the Page-Wootters construction across different pictures, states of the universe and system-clock interactions. *Physical Review D* 108, 063507.
- Rignon-Bret, A., and Elouard, C. (2026). Quantum stochastic thermodynamics of macroscopic systems: an algebraic approach. arXiv:2607.09242v1, preprint.
- Ryu, S., and Takayanagi, T. (2006). Holographic Derivation of Entanglement Entropy from AdS/CFT. *Physical Review Letters* 96, 181602.
- Safranek, D., Deutsch, J. M., and Aguirre, A. (2019). Quantum coarse-grained entropy and thermalization in closed systems. *Physical Review A* 99, 012103.
- Shaari, J. S. (2026). Informational arrow of time in an extended two-qubit Page-Wooters model. *Physics Letters A* 577, 131456.
- Shokrian Zini, M., Brown, A. R., and Freedman, M. (2023). The Smallest Interacting Universe. *Journal of High Energy Physics* 2023, 82.
- Stoica, O. C. (2026). The clock ambiguity problem: extended or extinguished?. arXiv:2604.21805.
- Styliaris, G., Anand, N., and Zanardi, P. (2021). Information Scrambling over Bipartitions: Equilibration, Entropy Production, and Typicality. *Physical Review Letters* 126, 030601.
- Szalay, S., Zimboras, Z., Mate, M., Barcza, G., Schilling, C., and Legeza, O. (2021). Fermionic systems for quantum information people. *Journal of Physics A: Mathematical and Theoretical* 54, 393001.

- Teixido-Bonfill, A., Schindler, J., and Safranek, D. (2025). Entropic partial orderings of quantum measurements. *Physica Scripta*. DOI: 10.1088/1402-4896/ad977c.
- Thirring, W., Bertlmann, R. A., Kohler, P., and Narnhofer, H. (2011). Entanglement or separability: The choice of how to factorize the algebra of a density matrix. *European Physical Journal D* 64, 181-196.
- Van Raamsdonk, M. (2010). Building up spacetime with quantum entanglement. *General Relativity and Gravitation* 42, 2323-2329.
- Wiseman, H. M., and Vaccaro, J. A. (2003). The entanglement of indistinguishable particles shared between two parties. *Physical Review Letters* 91, 097902.
- Witten, E. (2018). Notes on Some Entanglement Properties of Quantum Field Theory. *Reviews of Modern Physics* 90, 045003.
- Zanardi, P. (2001). Virtual Quantum Subsystems. *Physical Review Letters* 87, 077901.
- Zanardi, P. (2026). Universal holonomic control of algebras of observables. arXiv:2609.28662v1, preprint.
- Zanardi, P., Dallas, E., Andreadakis, F., and Lloyd, S. (2024). Operational Quantum Mereology and Minimal Scrambling. *Quantum* 8, 1406.

Conceptual Provenance and AI-Assistance Disclosure

The originating cosmological intuition is the author's. It developed from questions about Maxwell's demon, statistical entropy, heat death, and whether the apparent end and beginning of a cosmic cycle must describe different universes at all. In particular, the author proposed that the universe remains one and the same global informational reality through the putative cycle, while the apparent succession of an end and a beginning reflects how that information is organized or described. The later formulation in terms of local readability and physically privileged observable algebras was developed with LLM assistance. The informational interpretation of the dark sector also originated in the author's brainstorming, but is retained here only as a separate speculative extension.

Multiple large language models were used extensively during the subsequent development of the manuscript. Their contributions included criticism and counterarguments; development and revision of the operator-algebraic notation, the boundary ansatz, relational-time connections, the finite-dimensional illustrations, the EC/CAR candidate calculation, the finite-time selector study, the measurement-entropy and access analyses, and the finite entropy-stability and transport analyses; literature discovery and comparison; code drafting and review; manuscript organization; English drafting and rewriting; and preparation of publication metadata. Much of the present wording, formal presentation, and exploratory code was generated or revised with LLM assistance. Some AI outputs made incorrect or overstated claims and were removed or qualified during revision.

No independent expert validation is claimed. The author directed the revisions, selected the claims retained in this version, and accepts responsibility for the final text, citations, errors, and omissions. LLMs are not listed as authors because they cannot verify the work, consent to publication, or assume responsibility for it.